

2(a)	any two from: 1 volume of, liquid / (orange) juice, consumed ; 2 idea of: time between drinking liquid and time of test ; e.g. left for 40 minutes before testing 3 sex / gender / all females ; 4 colour of light / flickering increases in even steps / time intervals of increase are always the same / AW ; 5 (16) mg of alcohol in, drink / (orange) juice ; 6 time allowed for drinking (5 mins) ;	2	5 A concentration of alcohol
2(b)	any two from: <i>idea of</i> shows the spread about the mean / AW ; indicates the reliability of the data ; to put <u>error bars</u> on the graph / see if the standard deviations overlap / see if the differences are significant ;	2	R accuracy

Question	Answer	Marks	Guidance
2(c)	any three from: 1 the frequency at which flicker fusion occurs varies greatly between individuals (with or without alcohol) ; 2 the frequency at which flicker fusion occurs is greater after alcohol consumption / on day 2 (in, most of the women / 9 out of 13 women) 3 there is little variation in the repeated measurements on day 1 or day 2 / result for individuals on day 1 or day 2 are consistent ; 4 it takes longer to notice the change from flickering to fused stimulus after alcohol consumption ; 5 (women) 4, 6 and 13, show little change / have lower CFFT, after alcohol ; 7 ref. to overlapping error bars for, some women / 4, 5, 6, 7, 8, 11, 12, 13, suggests the effect of alcohol is not significant or ref. to women with the highest flicker fusion (2 and 3) / 1, 2, 3, 9, do not have overlapping error bars so effect of alcohol is significant ; 8 ref. to a range of values from lowest to highest ; no alcohol: 17.3 ± 0.1 and 12.7 ± 0.1 alcohol: 18.5 ± 0.1 and 12.7 ± 0.1	3	

Question	Answer	Marks	Guidance
2(d)	<i>any three from:</i> very small sample / only 13 subjects ; all test subjects are female ; <i>idea of</i> habituation (as 10 tests carried out) ; not all age ranges included ; no indication of body mass ;	3	